MESSAGE QUEUES

Use case-
1.Sending notifications to users.
2.Processing of orders.
3.automations – for e.g. when we have to send the email marketing to multiples users, then in 2days again follow ups question/answers marketing etc. , then if user is interested then another email like that.
So, there is series of event that should happen. Which can be implemented by queue on FIFO basis.

4. Video processing -> let's say we have a 4k video ,
And we have to process it into different formats 360p , 470p, 720p , 1080p , 2156p , 4k . like this.

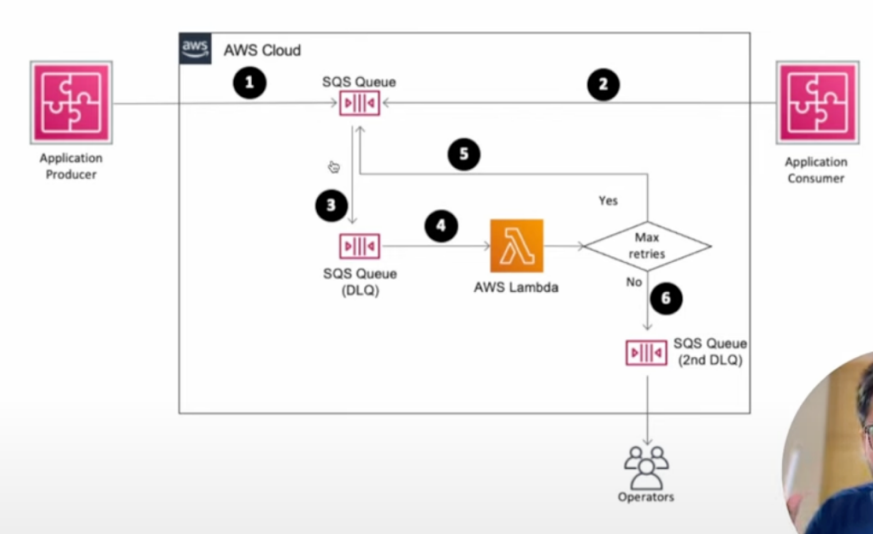
Practical Implementation-

Suppose, different users have uploaded 4k video, it gets processed in s3 bucket(temperory) ,
Suppose 10 videos are there to upload on YouTube , then the producer will push these 10videos in the queue , and we have a worker where it listens to the queue. If there is any job present in the queue , it will start processing it.
Now it depends how many workers we can setup , it depends on our traffic or load to the server.

Now , if while processing a video , if there is any error occurs , what we can do is we can , again enqueue it , so that the next time it will get processed.
That's fine if any network error or something happen ,

But what if that video/message is corrupted ? then it will get in infinite loop.

So , to solve this we have dead-letter-queue.
And this queue has different worker , and we will again try to process it , if it again fails then it means there is error in video/message.
And then push it into notification queue . showing error video could not be processed.



In BullMQ , we have two components.
1.queue
2.worker

And we have a Redis server

So first we have a Nodejs server , and we will make a queue in it.
We can give any name for the queue
And then we can push the jobs inside the queue.
And internally, queue works on redis. So, all the data inside queue is used/stored in redis. (redis server)

And next thing we will implement is worker ; we will define worker belongs to which queue. So the name has to be same as name of queue. And that worker is also connected to same redis server.

Since we r using BullMQ , we have option to set retryCount to retry in dead-letter-queue.